ALO.AI PRD V1.8.0 · WORKED EXAMPLE

Chidi's Alo.ai Session

One first session on a parent's phone, from sign-up to payment. Each step shows what is said and heard, what is on screen, which service does the work, what is sent where, and every number that gets calculated. The numbers come from running `aloai_scoring.py` on this example. Checked against PRD v1.8.0 (the 6-day hackathon build) and the Software Design Document.

Child: Chidi, 8, Primary 3, term 1 **Parent:** Ngozi, his mum

Where: home, Ngozi's Android phone, mobile data

Session: about 4.5 min of Chidi's time (standard profile, clips first)

Voice: one Google voice for Auntie Kemi (no pre-recorded audio)

day before

Before the session

Ngozi sets up an account once. Nothing about Chidi's reading exists yet.

Day −1

Ngozi signs up and gives consent

Ngozi · Web app · Account and Payments service · Firebase Auth · Firestore

ON SCREEN

> Sign-in: “Continue with Google”, or an email address and a sign-in link. Ngozi taps Google. Consent screen in plain English with a tick box: “I am the parent or guardian of this child.” A separate, unticked box asks about using recordings to improve the models. Then “Add a child”: first name, primary class, birth year.

SENT

`app → Firebase Auth`
Google sign-in; Firebase returns Ngozi's ID token, which the app sends with every request (no SMS, no password)

`Account service → Firestore`
consent record and child profile (the browser never writes to the database itself)

```
consents/c_81f2: {
  guardian_id: "g_44a1", child_id: "ch_9d0e",
  text_version: "1.0", attestation: true,
  sign_in_method: "google", verified_email: "ngozi@…",
training_opt_in: false,
  timestamp: "2026-10-13T18:02:11Z"
}
children/ch_9d0e: { guardian_id: "g_44a1", first_name: "
<encrypted>",
  primary_class: 3, birth_year: 2018, saved_tags: [] }
```

CALCULATED

- Starting level (PRD 6.7): first session, Primary 3, October is term 1 → 2 × 3 − 1 = **ARL 5**

0:00–0:32

Setup and story writing

While Chidi chats and counts, the system writes and checks his story in the background.

0:00

Ngozi taps Start reading check

Ngozi · Web app · Account and Payments service · Session Agent · Firestore · Cloud Storage

ON SCREEN

> “Find a quiet spot. Let Chidi hold the phone. Please don't help him read.” Big button: Start.

SENT

`app → Account service`

create session: checks the consent record and that this is the free first check, then returns a LiveKit room token and starts a Session Agent in that room

```
POST /v1/sessions { child_id: "ch_9d0e", context: "HOME" }
```

`Account service → Firestore`

creates the session

```
sessions/s_7b3c: { child_id: "ch_9d0e", context: "HOME",
  start_arl: 5, passage_source: "GENERATED", status: "SETUP" }
```

`Session Agent → Firestore`

checks the consent record again as it joins the room (no consent, no microphone)

`Session Agent → app`

links to pre-cache audio: Auntie Kemi's fixed prompts (made once per voice profile, vp_google_v1) and the ARL 5 listening story with its 2 questions (pre-made audio), about 0.4 MB as 32 kbps MP3. A bank fallback passage's clips would download only if the story failed its checks.

0:05

Interest chat

Chidi · Web app · Session Agent · Chirp streaming · Passage Generator (Core) · Cloud Storage · Firestore

HEARD / SAID

Auntie Kemi (Chirp 3 HD clip) *"Hello! I'm Auntie Kemi. Before we read, tell me: what kind of stories do you like?"*

Chidi *"I like football. And goats. My uncle has a goat in Ibadan."*

ON SCREEN

> Auntie Kemi's face with a pulsing ring while Chidi talks.

SENT

`app → Session Agent`
Chidi's audio over WebRTC (Opus, 16 kHz)

`Session Agent → Cloud Storage`
s_7b3c/interest/turn1.flac

`Session Agent → Gemini Flash`
the transcript from Chirp streaming, only to map it to topic tags

`Session Agent → Passage Generator`
only the tags, the safe detail and the target level

```
POST /v1/passages:generate { session_id: "s_7b3c", target_arl: 5,
  tags: ["football", "animals"], detail: "goat",
  voice_profile_id: "vp_google_v1", length_profile: "STANDARD",
  deadline_ms: 20000 }
```

`Session Agent → Firestore`
tags saved to Chidi's profile (Ngozi consented)

```
children/ch_9d0e.saved_tags: ["football", "animals"]
```

CALCULATED

- The Session Agent's tagger (Gemini Flash) maps the answer to the fixed taxonomy: **football, animals**; safe detail **goat**
- **Dropped:** "my uncle" and "Ibadan". They never reach the story writer.

0:15

The story is written

Passage Generator (Core)

SENT

`Passage Generator → Gemini (Vertex AI)`
structured generation request; no name, no transcript, nothing about Chidi

```
{
  target_arl: 5, genre: "NARRATIVE", topic_tags:
["football","animals"],
  detail: "goat",
  blueprint: { words: [120,150], max_mean_sentence_len: 11,
               min_share_zipf_3_5: 0.90, max_share_3plus_syllables:
0.05 },
  names_allowed: ["Tunde","Mama Bisi","Ada","Bayo", "…"],
  template: "character, setting, problem, 2-3 events, resolution",
  items: { literal: 2, inferential: 1 }, spelling: "British"
}
```

`Gemini → Passage Generator`

title, 14 sentences, 10 propositions, 3 questions with rubrics and example answers (attempt 1, 11.6 s)

CALCULATED

- Code (no LLM) builds 128 tokens with sentence numbers and phrase boundaries (punctuation plus a spaCy dependency parse), which the phrasing score uses later. Names carry their IPA from the curated list, for the voice.

0:15

Meanwhile: mic check

Chidi · Web app · Session Agent

HEARD / SAID

Auntie Kemi *"Can you count from one to five for me?"*
Chirp 3 HD clip

Chidi *"One, two, three, four, five."*

Auntie Kemi *"Lovely! Your story is almost ready."*
Chirp 3 HD clip

ON SCREEN

> Auntie Kemi listening, with a small sound meter.

CALCULATED

- Web app: loudness −27 dBFS (needs above −42) **pass**; clipping 0% **pass**; connection 46 kbps **pass**
- About 10 s. There are no warm-up words in this build; they return with adult-voice detection in the pilot (PRD 5.6).

0:27

The generation gate checks the story

Passage Generator (Core) · Cloud Storage

CHECKED

CHECK	RESULT
Words	128
Mean sentence length	9.1 words
Common words (Zipf ≥ 3.5)	100%
3+ syllable words	2.4% (anything, Saturday, oranges)
Estimated level	ARL 5.2 (illustrative)
Safety	no issues
Names	Tunde, Mama Bisi
Inference is real (string check)	the answer spans sentences 7 and 8; no single sentence holds it
Critic A, given the passage	6 / 6

CHECK	RESULT
Critic B, never given the story	`1 / 4 on literal items`
Propositions	`10, 5 core`

SENT

`Passage Generator → Cloud Storage`
the shared clip store (gs://aloai-clips-eu/vp_google_v1/1/words/) already holds “That word is X.” for 80 of the passage's 88 distinct words (illustrative), so only 8 new clips are made, plus 3 question clips under p_gen_5a21/. Names are spoken from the curated IPA list (Tunde → 'tundɛ).

`Passage Generator → Cloud TTS`
synchronous requests for the 11 new clips, with custom pronunciations; about 1 s in parallel

```
{ voice: "en-GB-Chirp3-HD-<Name>", speaking_rate: 0.9,
  text: "That word is Tunde.",
  custom_pronunciations: [{ phrase: "Tunde", ipa: "'tundɛ" }] }
```

`Passage Generator → Firestore`
passage p_gen_5a21 stored permanently (text, features, gate results, model and prompt version)

`app ← Cloud Storage`
downloads the passage's clips (about 0.5 MB as 32 kbps MP3; about 0.9 MB for the whole session). Ready at 0:30.

NOTES

- Counts and frequencies above are real values for this passage (measured with the `wordfreq` library). The level estimate is illustrative (rule-based estimator).
- Passed on the first attempt; one retry was allowed before the bank fallback.

0:32–1:53

Reading aloud

Chidi reads the story on screen. Fluency is scored on CPU while he retells and answers.

0:32

Chidi reads the story aloud

Chidi · Web app · Session Agent · Chirp streaming · Cloud Storage

HEARD / SAID

Auntie Kemi Chirp 3 HD clip *“Here's a story just for you. Read it out loud. Start whenever you're ready.”*

ON SCREEN

“The Goat on the Football Field” in large type. No timer, no highlighting, no red marks. What Chidi actually read is marked below (hover a word):

> Tunde loved football more than anything. Every Saturday, he played with his friends on the field behind the market.

One morning, they heard a loud noise. A small brown goat ran onto the field! It chased the ball and kicked it with its ~~back~~ legs. The boys laughed and tried to catch it, but the goat was too fast for them. Then Tunde had an idea. He took a piece of bread from his bag and held it out. The goat stopped and walked slowly towards him. Tunde gave it the bread and tied a rope around its neck. Just then, Mama Bisi came running down the road. “My goat!” she shouted. “Thank you, Tunde!” She gave him two oranges, and the boys went back to their game.

told · substitution / omission · self-corrected · accent: both transcripts agree, correct

SENT

`app → Session Agent`
live audio over WebRTC

`Session Agent → Cloud Storage`
s_7b3c/phase1/reading.flac (the whole reading)

`Session Agent → Chirp 3 streaming`
the same audio; partial transcripts come back about every 300 ms

CALCULATED

- Position tracker fuzzy-matches the partial transcripts to the passage to know which word Chidi is on. This is only for live prompting; scoring happens afterwards.

0:54

Chidi gets stuck on “chased”

Chidi · Session Agent · Web app

HEARD / SAID

Auntie Kemi *“That word is chased.”*
Chirp 3 HD clip

SENT

`Session Agent → app`
play the stored clip “That word is chased.” (always a clip, in either voice mode)

`Session Agent (log)`
TOLD at token 35, 0:54.4

CALCULATED

- Trigger: the tracker says the next word is token 35, and LiveKit's voice-activity plugin shows 3.0 s of silence

1:53

Reading ends; fluency is scored

Session Agent · Reading Scorer · Cloud Storage · Firestore

SENT

`Session Agent → Pub/Sub`
phase1.recorded for s_7b3c

`Reading Scorer ← Cloud Storage`
reading.flac and the passage artifact

`Reading Scorer → Chirp 3 and Gemini Flash`
the reading audio, for two independent verbatim transcripts

`Reading Scorer → Firestore`
sessions/s_7b3c/phase1: the Phase 1 telemetry about 15 s later (2:08), while Chidi retells; then the event phase1.scored, which Core uses to assemble the report

```
{ passage_source: "GENERATED", words_attempted: 128,
  reading_seconds: 80.0, words_correct: 125,
  error_counts: { substitution: 1, omission: 1, told: 1,
mispronunciation: 0 },
  self_correction_count: 1, uncertain_count: 0,
  accuracy_pct: 97.7, wcpm: 93.8, self_correction_ratio: 0.33,
  prosody: { reported: true, pause_intrusion_rate: 0.154,
             phrasing: 38, expression: null, overall: 38 },
  flags: [] }
```

CALCULATED

- Inside the Reading Scorer (CPU, no GPU): two transcripts, each aligned to the passage → forced alignment gives each of the 128 words a start and end time → a word is labelled only where both transcripts agree
- `accuracy = (128 − 3) / 128 = 97.7%`
- `WCPM = 60 × (128 − 3) / 80 s = 93.8`
- `self-correction ratio = 1 / (1 + 1 sub + 1 omit) = 0.33` (told words excluded)
- Both transcripts wrote “the”, “them” and “then” where Chidi said /də/, /dɛm/, /dɛn/, so those words **count as correct**; 0 words were left out as uncertain. Checking pronunciation sound by sound, with a Nigerian English allowlist, comes with the pilot.
- `phrasing = 100 × (1 − 0.154 / 0.25) = 38` from 4 intrusive pauses in 26 phrases (pauses from Silero VAD, phrase boundaries from the spaCy parse stored with the passage). Expression is **null**: it's switched off until there's a Nigerian English reference voice to compare against.

1:55–3:15

Retell and questions

The text disappears. Each answer is recorded as its own audio file.

1:55

Retell

Chidi · Web app · Session Agent · Cloud Storage

HEARD / SAID

Auntie Kemi — Chirp 3 HD clip
“Well done! In your own words, tell me everything you remember from the story.”

Chidi — 31 s
“Tunde is playing football and a goat came, and the goat kick the ball. And they laugh. And the woman came and she gave him orange.”

Auntie Kemi *"Anything else?"*
Chirp 3 HD clip

Chidi *"No."*

ON SCREEN

> The story is gone. Auntie Kemi only.

SENT

`Session Agent → Cloud Storage`
s_7b3c/phase2/retell.flac

`Session Agent → Grader`
the retell's live transcript to the quick endpoint; quick proposition credits come back (weighted 6.5 of 15)

```
POST /v1/grade:quick { session_id: "s_7b3c", turn_id: "retell",
  kind: "RETELL", transcript: "Tunde is playing football and a goat
came…" }
```

CALCULATED

- The recording, not this live transcript, is what the full grading uses after the session

2:37

Three questions

Chidi · Web app · Session Agent · Gemini Live · Grader (Core) · Cloud Storage

HEARD / SAID

Auntie Kemi *"Where did Tunde and his friends play football?"*
Chirp 3 HD clip

Chidi *"On the field."*

Auntie Kemi *"What did Tunde give the goat?"*
Chirp 3 HD clip

Chidi *"Bread."*

Auntie Kemi *"Why did the goat stop running?"*
Chirp 3 HD clip

Chidi *"Because it was tired."*

SENT

`Session Agent → app`
play the pre-made question clip, e.g. p_gen_5a21/Q3.ogg

`Gemini Live → Session Agent`
tool call when Chidi finishes each answer

```
end_turn({ reason: "ANSWERED" })
```

`Session Agent → Cloud Storage`
one file per answer: q1.flac, q2.flac, q3.flac

`Session Agent → Grader`
each answer to the quick endpoint (POST /v1/grade:quick): one Flash pass, back in about 1.5 s (PRD 7.4)

CALCULATED

- Quick scores against the rubrics: Q1 **1** (rubric's 2 points needs "behind the market"), Q2 **2**, Q3 **0**

One voice: Gemini Live is set to the same voice name as the clips. If Chidi had said "say it again", Live would call `request_repeat` and the agent would replay the clip. Live itself never says question content.

3:12

Decide whether to run the listening story

Session Agent · Grader (Core) · aloai_scoring.py

HEARD / SAID

Auntie Kemi *"Now just listen. I'm going to read you a short story."*
Chirp 3 HD clip

CALCULATED

- The quick endpoint returns this **quick** RC once the retell and all three answers are scored (PRD 7.4); the final RC is computed after the session
- `retell = min(1, 6.5 / (0.6 × 15)) = 0.72` from the quick proposition credits
- `questions = (1 + 2 + 0) / 6 = 0.50`
- `RC = 100 × (0.4 × 0.72 + 0.6 × 0.50) = 58.9`
- `58.9 < 78` → run the listening module

3:15–4:42

Listening story

Runs because Chidi's reading comprehension came out low. This separates 'can't read the words' from 'doesn't understand the language'.

3:15

Listening story, retell and two questions

Chidi · Web app · Session Agent · Cloud Storage · Grader (Core)

HEARD / SAID

Auntie Kemi *"[Bank story L5-L02, "Ada's Umbrella": Ada forgets her umbrella at school, it rains, and her friend Bayo shares his on the walk home.]"*
Chirp 3 HD clip, 41 s

Chidi *"Ada forget umbrella. And rain is falling. And she go home with Bayo."*
28 s retell

Auntie Kemi *"What did Ada forget at school?"*
Chirp 3 HD clip

Chidi *"Her umbrella."*

Auntie Kemi *"Why did Ada's mother smile when she saw Bayo at the door?"*
Chirp 3 HD clip

Chidi *"Because she like him."*

ON SCREEN

> A simple picture of rain. No text.

SENT

`app`
plays the pre-cached listening-story audio (no network needed)

`Session Agent → Cloud Storage`
listen_retell.flac, l1.flac, l2.flac

CALCULATED

- This story is the same for every child at ARL 5, which makes it the fixed anchor for the profile decision

4:42

Goodbye

Chidi · Web app · Session Agent

HEARD / SAID — **Auntie Kemi** *"You did a great job today! See you next time."*
Chirp 3 HD clip

ON SCREEN

> Stars animation. No scores.

SENT

`Session Agent → Pub/Sub`
session.done for s_7b3c

4:42–5:25 +

After the session

Full grading, the arithmetic, the report, the Reading Coach's plan, and the upgrade.

4:43

Full grading

Grader (Core) · Cloud Storage · Firestore

SENT

`Grader → response-track ASR`
the 8 answer and retell files

`Grader → Gemini 2.5 Pro`
pass A and pass B in parallel, temperature 0, example answers shuffled

```
{ passage, propositions, items + rubrics + exemplars,
  retell_transcript: "Tunde is playing football and a goat came,
    and the goat [?kick?] the ball. And they laugh. And the woman
    came and she gave him orange.",
  answers: { Q1: "On the field.", Q2: "Bread.", Q3: "Because it was
tired." },
  primary_class: 3 }
```

`Gemini Pro → Grader`
GraderOutput per passage

```
propositions: P1 0.5, P2 0, P3 1, P4 1, P5 0.5, P6 0,
              P7 0, P8 0, P9 0.5, P10 1     (weighted credit 6.5 of
15)
items: Q1 1, Q2 2, Q3 0
miscue_tags: token 69 "piece"→"place" MEANING_DISRUPTING
needs_review: false
```

CALCULATED

- Items: pass A and pass B agree on all three, so no third pass
- Retell: pass A 6.5, pass B 6.5 → within 1.0, so the mean, **6.5**, is used (PRD 7.2)
- Quick vs final: all 3 items match (agreement 100%). Final RC 58.9 < 78 and the listening story ran, so the listening result is used (PRD 6.5)

- Response-track ASR and Gemini Live's own transcript agree, so no ASR_DISAGREEMENT flag
- Listening story: weighted credit 4.0 of 12; L1 2, L2 0

4:58

The code computes everything else

aloai_scoring.py · Firestore

CALCULATED

QUANTITY	CALCULATION
Word reading OK	`97.7% ≥ 95 and 93.8 ≥ 79 (P3 cut)`
Reading comprehension	`100 × (0.4 × 0.722 + 0.6 × 0.500)`
Listening comprehension	`100 × (0.4 × min(1, 4.0 / 7.2) + 0.6 ×`
Profile	RC < 70, LC < 70, word reading OK
Provisional?	|58.9 − 70| = 11.1 and |52.2 − 70| = 17.8, both ≥ 8
Accuracy band	`97.7% is 95–97.9`
Comprehension band	`58.9 is 50–69`
Overall band	the lower of the two
Reported level	ARL 5 − 1
Next passage	ARL 5 − 1
Book band	[L − 1, L] for this profile

SENT

`code → Firestore`
reports/s_7b3c (the JSON in the report section below), written once grading and the telemetry are both in, inside one Firestore transaction

`Firestore → app`
real-time listener; the report screen opens at 5:04

`Grader → Pub/Sub`
report.ready for s_7b3c

5:05

The Reading Coach writes the plan and the spoken summary

Reading Coach (Core) · Passage Generator (Core) · Firestore · Account and Payments service · Email provider

SENT

`Reading Coach ← tools`
get_report (the report above) and get_history (first session, so no history)

Reading Coach ← tools
get_activity_library for this profile at ARL 4: picks 3 of about 30, e.g. stop every few sentences and ask “why?”, talk about new words, read aloud together and discuss

Reading Coach ← tools
search_books for ARL 3–4, rich narrative, football and animals: 3 open-licence titles

Reading Coach → Passage Generator
queue_story: the next story, one saved interest plus one new topic

```
queue_story({ child_id: "ch_9d0e", arl: 4, tags: ["football",
"market"] })
  → p_gen_6c07, same gate as before
```

Reading Coach → Firestore
schedule_session in 14 days (the result isn’t provisional) and save_plan; parent_reporting filled in reports/s_7b3c

```
plans/pl_3e19: { child_id: "ch_9d0e", activity_ids:
["act_why_stop", "act_word_talk", "act_read_discuss"],
  book_ids: ["…3 titles…"], next_session_at: "2026-10-28",
template_used: false }
```

Reading Coach → Cloud TTS
the summary, spoken in Auntie Kemi’s voice for the report screen

Reading Coach → Account and Payments service → Email provider
send_parent_email, once: “Chidi’s reading report is ready” (no scores in the email)

CALCULATED

- Every tool call passes `authorize()` in `aloai_authz.py`: the scope is Chidi only, taken from a run context the server built, and `send_parent_email` is allowed once per run
- Validator (`validate_parent_summary` in `aloai_guardrails.py`): 57 words (limit 90) **pass**; the only number, 94, matches WCPM 93.8 rounded **pass**; no banned words **pass**
- Plan ready at 5:20, inside the 20 s target. Chidi’s next session skips the interest chat and starts reading about 15 s sooner.

5:25

Ngozi upgrades

Ngozi · Web app · Account and Payments service · Paystack · Firestore

ON SCREEN

> Ngozi taps “Get Chidi's weekly plan · ₦1,500/month”. Paystack's popup opens inside the app; she pays by card.

SENT

app → Account and Payments service
POST /v1/checkout { plan: "monthly" } with Ngozi’s sign-in token

Account and Payments service → Paystack
Initialize Transaction (the server sets the price)

```
{ email: "ngozi@…", amount: 150000, plan: "PLN_monthly_1500",
  channels: ["card"], metadata: { guardian_id: "g_44a1" } }
```

`Account and Payments service → app`
the access code; Paystack's popup opens with it

`Paystack → Account and Payments service`
webhooks charge.success and subscription.create; signature checked, each handled once, then answered 200

`Account and Payments service → Paystack`
Verify Transaction by reference before granting anything

`Account and Payments service → Firestore`
subscriptions/g_44a1 { status: "active", current_period_end: "2026-11-14" }

`Firestore → app`
the weekly plan unlocks

CALCULATED

- Checks: x-paystack-signature is the HMAC-SHA512 of the raw body; the event reference hasn't been processed before
- Paystack receives Ngozi's email and the amount. It receives nothing about Chidi.

What Ngozi sees, and the numbers behind it

The report screen opens about 20 seconds after the session ends. The left side is the parent view. The right side is the report JSON the code produced, which the Reading Coach and the next session also read. The spoken summary and the plan arrive about 15 seconds after the report, once the Reading Coach has written them and the code has checked them.

READING CHECK · 14 OCT

Chidi's reading

Played aloud in Auntie Kemi's voice: “Chidi read 94 words correctly in a minute and got almost every word right. Reading the words is a real strength. He remembered parts of the story but found it hard to explain why things happened, even when I read a story to him. This was a short check, so think of it as a first look.”

Words right	**98%**
Words correct per minute	**94**
Reading level	**Level 4 (about Primary 2)**
Working on	**Understanding stories**

Try this week: while you read together, stop every few sentences and ask "Why do you think that happened?" Let Chidi answer before you read on.

Get Chidi's weekly plan · ₦1,500/month


```
{
  "passage":   { "source": "GENERATED", "title":
"The Goat on the Football Field",
                 "topic_tags": ["football",
"animals"] },
  "fluency":   { "accuracy_pct": 97.7, "wcpm":
93.8, "words_attempted": 128,
                 "reading_seconds": 80,
"self_correction_ratio": 0.33,
                 "word_reading_ok": true },
  "prosody":   { "reported": true, "phrasing":
38, "expression": null, "overall": 38 },  //
expression off in v1.4
  "comprehension": { "reading_comprehension":
58.9, "retell_coverage": 0.72,
                 "item_score": 0.50,
"quick_reading_comprehension": 58.9,
                 "listening_module_run": true,
"listening_comprehension": 52.2,
                 "listening_used": true,
                 "quick_vs_final_item_agreement":
1.0 },
  "level":     { "passage_arl": 5, "band":
"SUPPORTED", "reported_arl": 4,
                 "at_least": false, "confirmed":
false, "next_passage_arl": 4,
                 "approx_primary_class": "Primary
2 (approximate, not normed)" },
  "profile":   { "code":
"FLUENT_NEEDS_LANGUAGE_SUPPORT", "provisional":
false },
  "recommendations": { "arl_min": 3, "arl_max":
4, "book_ids": ["…3 titles…"] },
  "flags":     [],
  "parent_reporting": { "summary_script": "Chidi
read 94 words correctly…",
                 "home_practice": "Stop every few
sentences and ask why…" },  // written by the
Reading Coach
  "context": "HOME", "capture_mode": "LIVE",
  "session_settings": { "voice_mode":
"cache_first", "session_profile": "standard" }
}
```


Not shown to Ngozi: the profile code, percentiles, the comprehension scores as numbers, or the word "struggling". **Never shown to Chidi:** any score at all.

What each service receives

The same session from the point of view of each service. The "never receives" column is the privacy design.

SERVICE	RECEIVES	PRODUCES	NEVER RECEIVES
Web app (Ngozi's phone)	Chidi's voice from the mic; pre-cached clips; the report	Audio stream; consent; taps	—
Firebase Auth	Ngozi’s Google sign-in	Her ID token	Anything about Chidi
Account and Payments service	Ngozi’s ID token; consent; the child profile; the session request; the Coach’s email request; Paystack webhooks	Consent record; session and LiveKit token; the email; the subscription	Any audio; card details (Paystack holds them)
Session Agent	Live audio; passage artifact; quick scores	Turn audio files; TOLD log; flow decisions	Ngozi's payment details
Gemini Live	Chidi's live audio, for turn-taking	end_turn and request_repeat calls; approved short replies; its own transcript (kept for comparison)	Question text; Chidi's name or profile; scores
Cloud TTS (Chirp 3 HD)	Passage words, questions and sentences with name pronunciations; the parent summary text, which includes Chidi's first name and results	Audio clips	Any audio of Chidi
Passage Generator (Core)	Topic tags, a safe detail, target level	The passage, its questions, audio clips	Chidi's name; his own words; anything he said about his family or town
Reading Scorer	The reading audio and the passage	Two transcripts, word labels, fluency, phrasing	Name, age, consent records
Chirp 3 and Gemini Flash (transcription)	The reading audio, for a verbatim transcript	One transcript each	Chidi's name, age or profile

SERVICE	RECEIVES	PRODUCES	NEVER RECEIVES
Grader (Core, Gemini Pro)	Passage, rubrics, answer transcripts, primary class	Proposition credits, item points, miscue tags	Name; audio of the reading
Reading Coach (Core)	Chidi’s report and history, through tools, for this one child	The plan, the parent summary, the next-story request, one email request	Other children’s data; payment details; any tool that can change a score
aloai_scoring.py	Telemetry + grader output	Every score, band, level, profile and recommendation	— (pure code, no network)
Paystack	Ngozi's email, amount, plan	Payment and subscription events	Anything about Chidi
Email provider	From the Account and Payments service: Ngozi's email and a “report ready” notice	—	Any scores or recordings

Gaps this walkthrough found, and how they were fixed

Tracing one real session exposed five things the PRD didn't specify. All five were fixed: one in v1.4 and four in v1.5. For the 6-day build, v1.8.0 then cut the read-back and deferred adult-voice detection to the pilot; their rules stay in `aloai_scoring.py` for when they return.

1. In-session quick scoring `Fixed in v1.5 · PRD 7.4, 6.5`

The Grader has a quick endpoint: one Flash pass per answer, about 1–2 s, called by the Session Agent. It drives only the listening decision (it also drove the read-back until v1.8.0 cut it). Final scores always come from the full two-pass grading. If they disagree: listening ran but final RC is 78 or more → result stored, not used (LC_NOT_USED); listening didn't run but final RC is under 70 → INSUFFICIENT_EVIDENCE (LC_MISSED) and the next session starts with the listening story.

2. “Struggled sentence” defined `Fixed in v1.5 · PRD 6.1`

Any sentence with a substitution, omission, told word or mispronunciation that wasn't self-corrected, or a pause over 1.5 s before a word. Accent variants and self-corrections don't count. Removed with the read-back in v1.8.0; the rule stays in the code for when it returns.

3. First-session starting level `Fixed in v1.5 · PRD 6.7`

ARL = 2 × class − 1 in term 1, 2 × class in terms 2 and 3; returning children continue from their last next-passage level. Term 1 is September–December by default, and schools can set their own calendar. Parents confirm the child's class each September.

4. Adult-voice detection `Fixed in v1.5 · PRD 5.6, 7.2`

The mic check and warm-up give about 8 s of the child's voice. Reading segments that don't match it and sit below 165 Hz count as adult speech; more than 2 s sets ASSISTED. Deferred to the pilot in v1.8.0, with the warm-up. Also fixed: when two grading passes give retell credit within 1.0 of each other, the mean is used.

5. Auntie Kemi had two voices `Fixed in v1.4 · PRD 4.6`

Every scripted line is a Chirp 3 HD clip, and Gemini Live uses the same voice name for unscripted replies.